

Java Statements

The core instructions that make up your program.

1. What Are Statements?

In Java, a computer program is a sequence of **statements**. Each statement is a complete instruction that tells the computer to perform a specific action.



A statement is like a sentence in English. Just as a sentence ends with a period (.), a Java statement ends with a semicolon (;).

Analogy:

2. The Mandatory Semicolon (;)

The semicolon is non-negotiable. It signals to the Java compiler that one instruction has finished and the next can begin.

✓ **CORRECT**

```
System.out.println("Hello");
```

✗ **INCORRECT**

```
System.out.println("Hello")
```

(Will cause a "';' expected" error)

3. Order of Execution

Java executes statements **sequentially**, which means from top to bottom, one after another.

- 1 Print World
- 2 Save User Age
- 3 Perform Math

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4. Common Types of Statements

Not all statements are for printing text. Here are the varieties you will encounter:

A. Variable Assignments

```
int score = 100; // Stores data in memory
```

B. Conditional (Decision) Statements

```
if (score > 50) {  
    System.out.println("Pass");  
}
```

C. Loop Statements

```
for (int i = 0; i < 3; i++) {  
    System.out.println("Repeat");  
}
```

5. Statement Blocks { }

Braces are used to group multiple statements into a single unit called a **block**. These are used frequently in methods and classes.

6. Statements vs. Expressions

Expression: Produces a value (e.g., `5 + 5`). It's a phrase.

Statement: Performs an action (e.g., `int x = 5 + 5;`). It's a complete sentence.

7. Common Beginner Pitfalls

- **Merging Statements:** Trying to write two instructions on one line without a semicolon in between.
- **Extra Semicolons:** Putting a semicolon after an `if` condition:
`if (x > 5); { ... }` ← The semicolon here ends the logic prematurely!

Final Takeaway

- Every program is a list of actions (Statements).
- Semicolons act as the signal for "Next Instruction Please!"
- Statements are read by the computer in the order you write them.
- Grouped statements live inside curly braces `{ }`.

Keep coding, keep building!